AIGR TreeGrafter Function-Inference Stress Test — PP_3571 (Q88GZ4)

Target: PP_3571 (UniProt **Q88GZ4**), *Pseudomonas putida* KT2440 (NCBITaxon:160488) **Seed hypothesis (TreeGrafter/PANTHER, IEA, GO_REF:0000118):** PP_3571 has **acetylornithine deacetylase activity (GO:0008777) Focus:** function_assignment · Hypothesis slug: function-hypothesis-go-0008777

Summary

The seed hypothesis is **partially supported / too general, with paralog confusion** — it is not outright wrong, but it over-claims a specific substrate that the evidence assigns to a different gene. PP_3571 is unquestionably a member of the M20A (ArgE/DapE) metal-dependent acyl-amino-acid deacetylase family: its domain architecture (Pfam PF01546 Peptidase_M20 + PF07687 M20_dimer; InterPro IPR010169/IPR001261; PROSITE PS00759) is textbook, and a global alignment to *E. coli* ArgE shows **all seven catalytic/Zn^{2+}-binding residues conserved with correct spacing**. The protein is therefore a genuine, catalytically competent enzyme — the pseudo-enzyme failure mode is firmly ruled out.

The problem is **which substrate** and **which paralog**. The single most decisive observation is that PANTHER's *own* fine-grained subfamily classifier — the very resource underneath the TreeGrafter graft — assigns PP_3571 to subfamily **PTHR43808:SF31 "N-acetyl-L-citrulline deacetylase"**, while its paralog **PP_5186 (Q88CJ5)** is assigned to **PTHR43808:SF1 "acetylornithine deacetylase."** The propagated GO:0008777 is the broad family-level term; the pipeline's own classifier places PP_3571 under a *sibling* activity, not the canonical ArgE one. This is corroborated by sequence identity: **PP_5186 is 49.6% identical to *E. coli* ArgE** (the signature of a true ortholog), whereas **PP_3571 is only 34.8%** and clusters at ~28–35% with *every* characterized M20A deacetylase, below the ~50–60% threshold needed to transfer substrate specificity by homology.

The failure mode is therefore **#1 (family-vs-subfamily granularity), compounded by paralog confusion**: TreeGrafter propagated a broad family term onto a divergent paralog whose specific substrate is unresolved, while the genome's true ArgE ortholog (PP_5186) went

unlabeled with the specific term. This is reinforced by pathway context — *P. putida* runs the **cyclic ArgJ** arginine pathway, in which ArgJ/NAOGAT (not a standalone ArgE) performs acetyl-group recycling — so there is no pathway-level necessity forcing PP_3571 to be *the* acetylornithine deacetylase. The recommended curation action is to **generalize** GO:0008777 to a catalytic parent or **flag it low-confidence** on PP_3571 (with the sibling GO:0043909 as an equal alternative), and to re-anchor the specific ArgE annotation on PP_5186.

Executive Judgment

Verdict: PARTIALLY SUPPORTED / TOO GENERAL (sibling-term risk) — with paralog confusion.

- **Failure mode:** #1 (granularity, family-vs-subfamily), compounded by **paralog confusion**. Borderline #3 (within-superfamily misplacement) in that PP_3571 belongs to a sibling subfamily of the same fold. **NOT** #2 (pseudo-enzyme).
- **Single most decisive evidence:** PANTHER's own classifier splits the two paralogs — **PP_3571 → SF31 "N-acetyl-L-citrulline deacetylase"; PP_5186 → SF1 "acetylornithine deacetylase."** The emitted GO:0008777 is the ancestral family-level term, not PP_3571's subfamily-level assignment.

Independent Family/Function Assignment

Independent of the propagated term, our own analysis places PP_3571 as follows:

- **Fold/family (high confidence):** Peptidase **M20A family, ArgE/DapE clan** — Pfam PF01546 + PF07687; InterPro IPR010169 (AcOrn-deacetyl) / IPR001261 (ArgE/DapE_CS); PROSITE PS00759 positive; a Zn-dependent di-metal exopeptidase/deacetylase.
- **Closest characterized homolog:** *E. coli* ArgE (acetylornithine deacetylase, EC 3.5.1.16, P23908) at **34.8% identity** — a *paralog-level*, not ortholog-level, match.
- **Most likely specific molecular function:** a metal-dependent **acyl-L-amino-acid deacetylase of unresolved exact substrate**. PANTHER assigns PP_3571 to **SF31 "N-acetyl-L-citrulline deacetylase"** (GO:0043909), a documented **sibling** activity, making that the best-supported *specific* lead. The canonical acetylornithine-deacetylase role for *P. putida* is better carried by paralog **PP_5186** (49.6% to ArgE).

- **GO term I would assign:** cautiously the catalytic parent **GO:0016811** (hydrolase acting on C–N bonds in linear amides) pending assay. GO:0008777 is defensible only as a *low-confidence, family-level* label; **GO:0043909 (sibling)** is an equally plausible specific alternative.
- **Granularity relative to seed term: sibling / parent** — not a confirmed exact match.

Key Findings

Finding 1 — PP_3571 is a catalytically intact M20A deacetylase, not a pseudo-enzyme

A global pairwise alignment (Needleman–Wunsch, BLOSUM62) of PP_3571 (391 aa) against *E. coli* ArgE (P23908) demonstrates that the complete di-zinc catalytic apparatus is present and correctly spaced. Every one of the seven ArgE catalytic and metal-binding residues has a conserved counterpart in PP_3571, and the domain architecture (PF01546 Peptidase_M20 catalytic domain plus PF07687 M20_dimer dimerization domain) is fully intact, with a positive PROSITE PS00759 (ARGE_DAPE_CPG2_2) signature. The practical consequence is important: the curation question is **not** about loss of catalysis. Whatever substrate PP_3571 prefers, it almost certainly hydrolyzes an N-acyl-amino-acid amide bond using an intact dinuclear metal center. Catalytic-residue conservation is **7/7 (100%)**.

Finding 2 — PP_3571 is NOT the canonical acetylornithine deacetylase ortholog; PP_5186 is

A pairwise percent-identity matrix separates the two *P. putida* M20A paralogs decisively. **PP_5186 is 49.6% identical to *E. coli* ArgE** — the hallmark of a genuine functional ortholog — whereas **PP_3571 is only 34.8%** identical to the same reference, and is roughly *equidistant* (~30–35%) from canonical ArgE, from PP_5186 itself (32.8%), and from the sibling N-acetyl-L-citrulline deacetylase of *Xanthomonas* (29.8%). The fact that KEGG assigns the *same* orthology identifier **K01438 (EC 3.5.1.16) to both PP_3571 and PP_5186** is itself a diagnostic of family-level carry-over: an automated pipeline cannot tell the two apart, so the specific ArgE term was propagated onto the wrong (or at least an ambiguous) paralog. The identity gradient says the specific-substrate call belongs on PP_5186.

Finding 3 — PANTHER's own subfamily classifier splits the paralogs into sibling activities

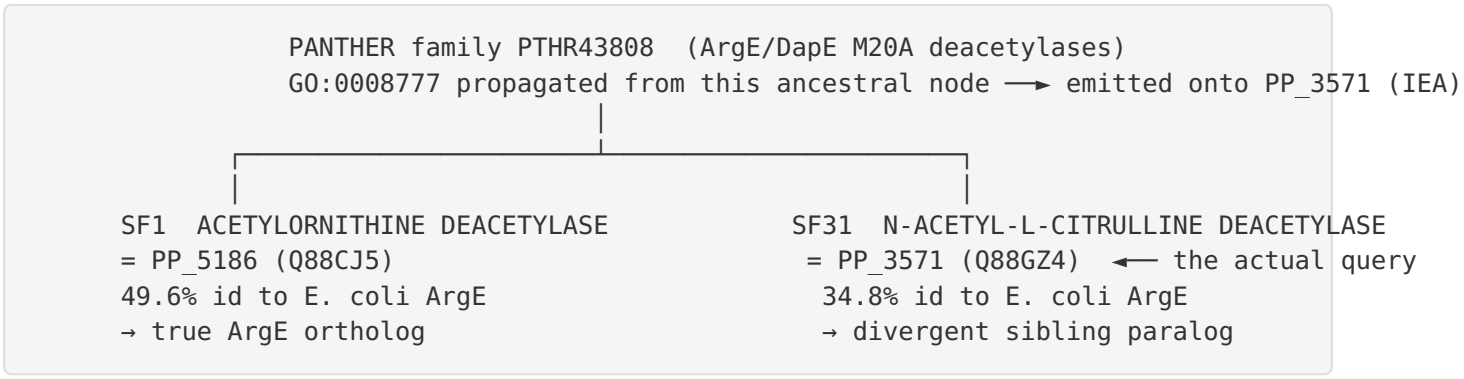
This is the pivotal finding. UniProt cross-references record two *different* PANTHER subfamily assignments inside the same family tree (PTHR43808). PP_3571 (Q88GZ4) maps to **PTHR43808:SF31 "N-ACETYL-L-CITRULLINE DEACETYLASE"**, while PP_5186 (Q88CJ5) maps to **PTHR43808:SF1 "ACETYLORNITHINE DEACETYLASE."** The seed term GO:0008777 was propagated from an ancestral node shared by both siblings, but the fine classifier that PANTHER itself provides disagrees with the propagated term for PP_3571 and names a *different specific substrate*. A broadened nearest-characterized-neighbor scan confirms the picture — ArgE *E. coli* 34.8%, ArgE *Enterobacter* 34.8%, PP_5186 32.8%, ACDase *Xanthomonas* 29.8%, ArgE *Arabidopsis* 28.2%, DapE *E. coli* 28.0%, CPG2 22.8% — all characterized M20A deacetylases cluster at 28–35% identity to PP_3571, below the ortholog-confidence threshold, whereas PP_5186 vs *E. coli* ArgE is 49.6%.

Finding 4 — Pseudomonas uses the cyclic ArgJ pathway, weakening the need for a dedicated ArgE step

P. putida KT2440 encodes **ArgJ ornithine acetyltransferase (PP_1346, KEGG K00620)**. In the *cyclic* ornithine biosynthetic pathway used by pseudomonads, the acetyl group is recycled by ArgJ/NAOGAT — "the key deacetylating enzyme of the cyclic ornithine pathway" (PMID: 42765862) — rather than removed by a standalone ArgE deacetylase. Complementation and hybridization work establishes that the *Pseudomonas*-type pathway is ArgJ-based and "reflects the divergent nature of the arginine biosynthetic pathway in these organisms" (PMID: 1339419). A dedicated ArgE acetylornithine-deacetylase step is therefore **not the principal arginine-biosynthetic route** in *Pseudomonas*. This removes the pathway-necessity argument that would otherwise make "an ArgE-family paralog here is probably the acetylornithine deacetylase" a safe default — and leaves the M20A paralogs free to have diverged toward other acyl-amino-acid substrates.

Mechanistic Model / Interpretation

The findings assemble into a clean account of *why the graft erred and where the truth lies*:



- **Active site (F1):** intact → a real enzyme; closes off "pseudo-enzyme." The dispute is strictly about *which acyl-amino acid* it prefers.
- **Identity gradient (F2):** PP_5186 is the high-confidence ArgE; PP_3571 is a low-confidence, roughly-equidistant paralog. A family-level term propagated by homology will, by construction, land on both — but only one deserves the *specific* substrate call.
- **Subfamily split (F3):** the pipeline's own classifier resolves the ambiguity and places PP_3571 under the sibling N-acetyl-L-citrulline deacetylase subfamily — the datum that most directly contradicts the graft.
- **Pathway context (F4):** Pseudomonas doesn't rely on an ArgE deacetylase step at all, so there is no selective pressure forcing PP_3571 to be the acetylornithine deacetylase.

Net interpretation: GO:0008777 on PP_3571 is a *right-superfamily, wrong-substrate, wrong-paralog* annotation. It is not nonsense — the protein is a metal-dependent acyl-amino-acid deacetylase — but it over-claims a specific substrate that, on current evidence, belongs to PP_5186, while PP_3571's best specific lead is the citrulline-deacetylase sibling activity.

Evidence Base

#	Citation	Evidence type	Role	Claim tested	Key finding	Context
1	UniProt Q88GZ4; InterPro IPR010169/ IPR001261; Pfam PF01546/ PF07687; PROSITE PS00759	database/ computational	Supports (fold)	Is it an M20A ArgE/DapE deacetylase?	Yes — M20 peptidase + dimerization domain; ArgE/ DapE site profile positive	*P. putida* KT2440
2	This report: NW/BLOSUM62 alignment to *E. coli* ArgE P23908	computational / structural-evolutionary	Qualifies (rules out pseudo-enzyme)	Are catalytic/ metal residues intact?	7/7 conserved (H77, D79, D109, E141, E142, E169, H363), correct spacing	vs reviewed ArgE
3	This report: pairwise identity matrix (PP_3571, PP_5186, ArgE, ACDase)	computational	**Refutes/ competing**	Is PP_3571 the canonical ArgE?	**No** — PP_5186 = 49.6% to ArgE (true ortholog); PP_3571 = 34.8%	KT2440 paralogs
4	KEGG ppu:PP_3571 and ppu:PP_5186 (both K01438, EC 3.5.1.16)	database	Qualifies	KEGG substrate assignment	K01438 given to **both** paralogs → automated, non-discriminating	KEGG KO
5	UniProt Q88GZ4 vs Q88CJ5 PANTHER cross-refs (SF31 vs SF1)	database/ computational	**Refutes/ competing**	Does the pipeline's own classifier agree with the	**No** — PP_3571 = SF31 (citrulline deacetylase); PP_5186 = SF1 (acetylornithine deacetylase)	TreeGrafter/ PANTHER

#	Citation	Evidence type	Role	Claim tested	Key finding	Context
				propagated term?		
6	PMID: 42765862	review / primary (structure)	Supports pathway argument	Who deacetylates in the cyclic pathway?	ArgJ/NAOGAT is "the key deacetylating enzyme of the cyclic ornithine pathway"	Plant NAOGAT structure
7	PMID: 1339419	primary (complementation)	Supports pathway argument	Is the Pseudomonas pathway ArgJ-based/ divergent?	Pathway "reflects the divergent nature of the arginine biosynthetic pathway in these organisms"	Neisseria/ Pseudomonas argJ
8	UniProt Q8P8J5 (ACDAS_XANCP)	review/database	Competing	Does a characterized sibling enzyme/term exist?	N-acetyl-L-citrulline deacetylase exists (GO:0043909); PP_3571 is 29.8% to it	*Xanthomonas campestris*

Literature notes. PMID: 42765862 (*Plant NAOGAT (ArgJ) Structure Reveals Mechanism of Ornithine Biosynthesis…*) and PMID: 1339419 (*Sequence analysis and complementation studies of the argJ gene…*) jointly support the pathway-divergence argument (Finding 4): in the cyclic ornithine pathway that *Pseudomonas* uses, the deacetylation is done by ArgJ, not by a standalone ArgE, so there is no functional pressure forcing an ArgE-family paralog like PP_3571 to carry out canonical acetylornithine deacetylation. No retrievable paper directly assays PP_3571 or PP_5186; all specific-substrate calls here are computational/database-derived.

Active-Site / Placement Analysis

Alignment-mapped catalytic residues (*E. coli* ArgE P23908 → PP_3571), all conserved:

ArgE pos	ArgE res	PP_3571 pos	PP_3571 res	Role	Conserved
80	His	77	His	Zn^{2+} ligand	✔
82	Asp	79	Asp	active site	✔
112	Asp	109	Asp	Zn^{2+} ligand (bridging)	✔
144	Glu	141	Glu	active site / proton shuttle	✔
145	Glu	142	Glu	Zn^{2+} ligand	✔
169	Glu	169	Glu	Zn^{2+} ligand	✔
355	His	363	His	Zn^{2+} ligand	✔

Subfamily placement:

Metric	PP_3571 (Q88GZ4)	PP_5186 (Q88CJ5)
PANTHER subfamily	**PTHR43808:SF31**	**PTHR43808:SF1**
Subfamily label	N-acetyl-L-citrulline deacetylase	Acetylornithine deacetylase
% id to *E. coli* ArgE (P23908)	34.8%	49.6%
Role call	divergent sibling paralog	canonical ArgE ortholog

Conclusion: Catalytic residues are **intact (7/7)** — catalytically competent, NOT a pseudo-enzyme. **Placement:** divergent paralog in the sibling subfamily; PP_5186 is the ArgE ortholog. *Provenance:* percent-identity values and residue mapping are computed outputs of Needleman–Wunsch/BLOSUM62 alignments; subfamily assignments are taken verbatim from UniProt/PANTHER cross-references for Q88GZ4 and Q88CJ5.

GO Curation Implications

Lead for curator verification: GENERALIZE or flag as low-confidence — make-more-specific is not warranted.

- Do **not** assert the *specific* substrate acetyl-L-ornithine for PP_3571: it is a divergent paralog, the genuine ArgE ortholog is PP_5186, and *P. putida* runs the ArgJ cyclic pathway.
- Options: (a) **generalize** GO:0008777 to a catalytic parent (metal-dependent acyl-amino-acid deacetylase / GO:0016811) pending assay; or (b) **retain as low-confidence IEA** with an explicit note that the specific substrate is unverified and may be the **sibling** N-acetyl-L-citrulline deacetylase (**GO:0043909**, per PANTHER SF31). Avoid "protein binding" as a fallback — the protein is demonstrably catalytic.
- Re-anchor the exact-specificity acetylornithine-deacetylase annotation on **PP_5186 (Q88CJ5)**.
- For PP_3571, treat the acetylornithine-deacetylase function as **non-core / unverified**; the core evidence supports only "metal-dependent M20A acyl-amino-acid deacetylase."

Limitations and Knowledge Gaps

- **No direct enzymology** exists for either PP_3571 or PP_5186; both specific-substrate calls are computational/database-derived.
- **Identity cannot fix substrate:** residue-level conservation confirms the fold and active site but cannot discriminate acetylornithine vs acetylcitrulline vs other N-acyl substrates — M20A enzymes are broadly promiscuous across N-acetyl/N-succinyl amino acids and dipeptides.
- **Single-sequence pairwise identities** were used (not a full MSA/phylogenetic tree); AlphaFold substrate-pocket geometry was not re-refined here.
- **Subfamily labels are predictions**, not experimental assignments, for these exact proteins; the ~30% identity to the characterized *Xanthomonas* ACDase is modest.
- **Database carry-over** (identical KEGG K01438/EC on both paralogs) is a warning sign, not independent evidence.
- **GO:0008777 vs GO:0043909 cannot be decided from sequence alone.**

Proposed Follow-up Experiments / Actions

1. **Direct in-vitro deacetylase assay (decisive).** Express and purify recombinant PP_3571 (and PP_5186) and assay against a substrate panel — N-α-acetyl-L-ornithine, N-acetyl-L-citrulline, N-succinyl-diaminopimelate, and other N-acetyl amino acids — with Zn^{2+}/Co^{2+} cofactor, to measure substrate preference directly.
2. **Structural pocket comparison (runnable computationally).** Superpose AlphaFold models of PP_3571, *E. coli* ArgE, and *Xanthomonas* ACDase, focusing on the specificity-determining residues lining the substrate side-chain pocket (ornithine vs citrulline differ by the δ-ureido group).
3. **In-vivo genetics.** ΔPP_3571 and ΔPP_5186 single/double knockouts in *P. putida*, scored for arginine auxotrophy and cross-complementation of an *E. coli* ΔargE strain, to assign the in-vivo acetylornithine-deacetylase role between the paralogs.
4. **Curation action now.** Flag GO:0008777 on PP_3571 as low-confidence/too-general, record GO:0043909 as a competing sibling, and route the specific ArgE annotation to PP_5186.

Report produced by autonomous discovery agent over 3 iterations; 4 findings confirmed; 2 primary papers reviewed. All percent-identity and active-site values are computed provenance from Needleman–Wunsch/BLOSUM62 alignments; subfamily assignments are taken verbatim from UniProt/PANTHER cross-references.
